

```
1  Library IEEE;
2  Use IEEE.std_logic_1164.all;
3
4  Entity LogicUnit is
5      Generic ( N : natural := 64 );
6      Port (
7          A      : in  std_logic_vector(N-1 downto 0);
8          B      : in  std_logic_vector(N-1 downto 0);
9          LogicFN : in  std_logic_vector(1 downto 0);
10         Y      : out std_logic_vector(N-1 downto 0)
11     );
12 End Entity LogicUnit;
13
14
15 Architecture Behavioral of LogicUnit is
16 Begin
17
18
19     process(A, B, LogicFN)
20     begin
21         case LogicFN is
22             when "00"    => Y <= B;
23             when "01"    => Y <= A xor B;
24             when "10"    => Y <= A or B;
25             when "11"    => Y <= A and B;
26             when others => Y <= (others => 'X');
27         End case;
28     End process;
29
30 End Architecture Behavioral;
```